

The LikhaKahoy Woodcrafts | IT105 – Database Systems

Group Members:

Benigay, Nicole S.

Cornal, Arleczar G.

Realubit, Aira Chelzei N.

Sola, Samantha Louise V.

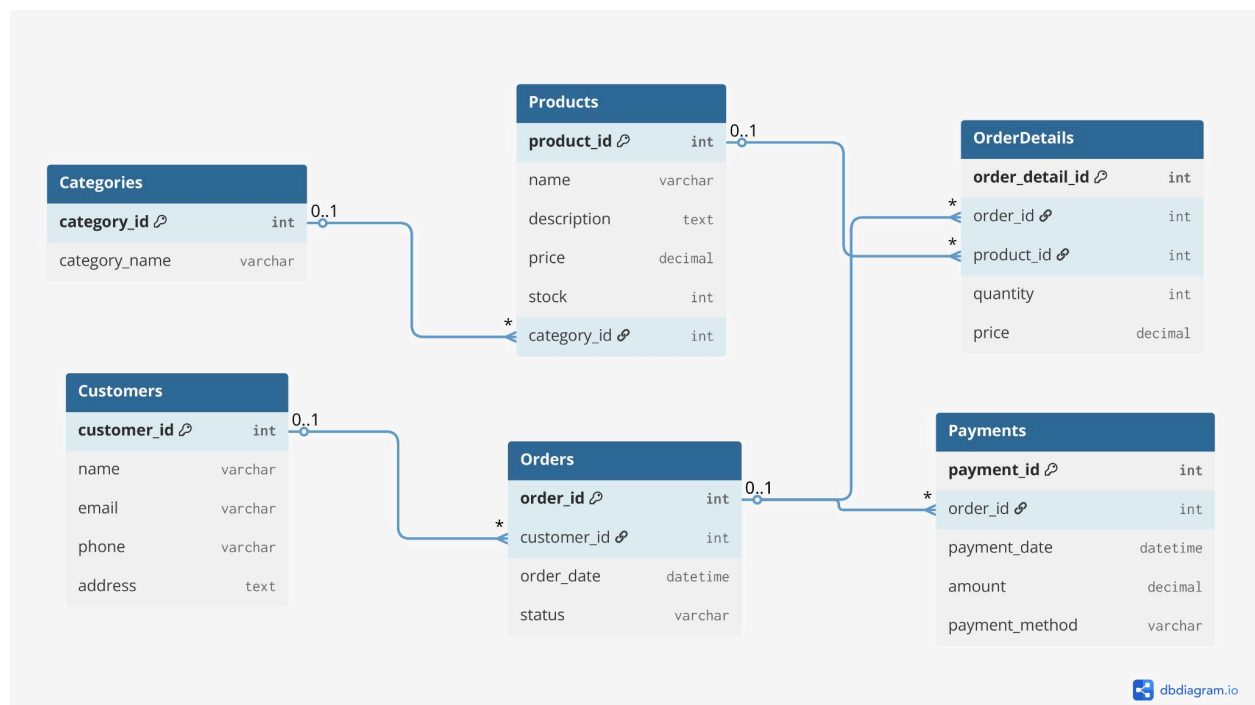
Tagle, Kerwin L.

System Description

The LikhaKahoy Woodcrafts project focuses on designing and developing a LikhaKahoy Woodcrafts Database System that stores, organizes, and manages essential business data. The system includes pro details, customer information, orders, payments, and sales records.

The database aims to improve data organization, ensure accuracy, and make information easily accessible for future system development. It serves as the foundation for a potential business management system for woodcraft businesses.

Entity Relationship Diagram (ERD)



Database Schema and Table Descriptions

Customers Table

Column	Type	Constraints	Description
customer_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the customer
Name	VARCHAR(50)	NOT NULL	Customer's full name
Email	VARCHAR(50)	NOT NULL, UNIQUE	Customer's email address
Phone	VARCHAR(20)	-	Contact number
Address	TEXT	-	Customer's address
Created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation date

Categories Table

Column	Type	Constraints	Description
category_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique category ID
category_name	VARCHAR(100)	NOT NULL	Name of the category

Products Table

Column	Type	Constraints	Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the customer
name	VARCHAR(100)	NOT NULL	Customer's full name
description	TEXT	-	Customer's email address
price	DECIMAL(10,2)	NOT NULL	Contact number
stock	INT	DEFAULT 0	Customer's address
category_id	INT	FOREIGN KEY → Categories(category_id), SET NULL	Record creation date
created_at	INT	DEFAULT CURRENT_TIMESTAMP	Record creation date

Orders Table

Column	Type	Constraints	Description
order_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique order ID
customer_id	INT	FOREIGN KEY → Customers(customer_id), CASCADE	Linked customer
order_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date of order
status	VARCHAR(50)	DEFAULT 'Pending'	Order status

OrderDetails Table

Column	Type	Constraints	Description
order_detail_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique order detail ID
order_id	INT	FOREIGN KEY → Customers(customer_id), CASCADE	Linked order
product_id	INT	FOREIGN KEY → Customers(customer_id), CASCADE	Linked product
quantity	INT	NOT NULL	Quantity ordered
price	Decimal(10,2)	NOT NULL	Price at the time of purchase

Payments Table

Column	Type	Constraints	Description
payment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique payment ID
order_id	INT	FOREIGN KEY → Customers(customer_id), CASCADE	Linked order
payment_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Payment date
amount	DECIMAL (10,2)	NOT NULL	Payment amount
payment_method	VARCHAR(50)	-	Mode of payment

Basic Query

```
SHOW TABLES;  
DESCRIBE Customers;  
DESCRIBE Categories;  
DESCRIBE Products;  
DESCRIBE Orders;  
DESCRIBE OrderDetails;  
DESCRIBE Payments;
```

Screenshots:

Customers Table

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
SELECT * FROM Customers;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

customer_id	name	email	phone	address	created_at
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Payment Table

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0029 seconds.)

```
SELECT o.order_id, c.name AS customer_name, o.status, p.amount, p.payment_method FROM Orders o JOIN Customers c ON o.customer_id = c.customer_id LEFT JOIN Payments p ON o.order_id = p.order_id;
```

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order_id	customer_name	status	amount	payment_method
----------	---------------	--------	--------	----------------

Product Table

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT * FROM Products;
```

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product_id	name	description	price	stock	category_id	created_at
------------	------	-------------	-------	-------	-------------	------------

Categories

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
SELECT * FROM `Categories` WHERE 1;
```

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category_id	category_name
-------------	---------------

Orders Table

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT o.order_id, c.name AS customer_name, o.order_date, o.status FROM Orders o JOIN Customers c ON o.customer_id = c.customer_id;
```

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order_id	customer_name	order_date	status
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BI Query

```
SELECT
  YEAR(payment_date) AS year,
  MONTH(payment_date) AS month,
  SUM(amount) AS monthly_revenue
FROM Payments
GROUP BY YEAR(payment_date), MONTH(payment_date)
ORDER BY year, month;
```

Screenshot:

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	year	month	monthly_revenue			
▶	2026	4	252115213.00			

Optimized

```
EXPLAIN
SELECT
  YEAR(payment_date) AS year,
  MONTH(payment_date) AS month,
  SUM(amount) AS monthly_revenue
FROM Payments
GROUP BY YEAR(payment_date), MONTH(payment_date)
ORDER BY year, month;
```

Screenshot:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	Payments	NULL	ALL	idx_payments_date	NULL	NULL	NULL	9908	100.00	Using temporary; Using filesort

Role Assignments & Contribution Summary

Name	Role	Contribution
Benigay, Nicole S.	Project Manager	Led the team by organizing tasks and setting clear goals; guided members throughout development; ensured effective communication and collaboration; addressed challenges and conflicts; kept the project on track; and ensured successful completion of the system within the given time.
Realubit, Aira Chelzei N.	Database Administrator (DBA)	Designed and managed the database structure; created tables, relationships, and constraints; ensured data accuracy and consistency; performed data validation and optimization; and supported system functionality by maintaining efficient database operations.
Cornal,Arleczar G.	SQL Developer	Developed and executed SQL queries for data retrieval and manipulation; created stored procedures, views, and functions; ensured data accuracy and consistency; optimized queries for better performance; and supported system functionality through efficient database operations.
Sola, Samantha Louise V.	QA/Tester	Tested the system to identify bugs and errors; created and executed test cases; verified system functionality and performance; documented issues and reported them to the development team; and ensured the system

		met quality standards before deployment.
Tagle, Kerwin L.	Documentation Lead	Managed and compiled all project documents; coordinated with team members to gather information; ensured completeness and accuracy; and supported clear communication through well-structured documentation.
Sola, Samantha Louise V.	Security Officer	Ensured system security by identifying potential vulnerabilities; implemented security measures such as user authentication and access control; monitored data protection; and helped prevent unauthorized access to the system.

Backup and Recovery Steps

Backup Command Used:

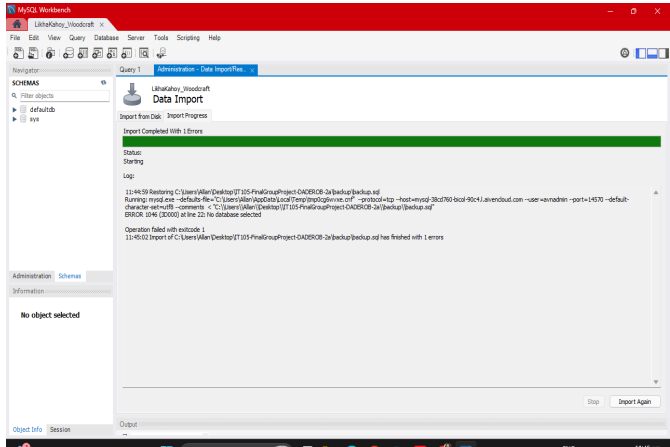
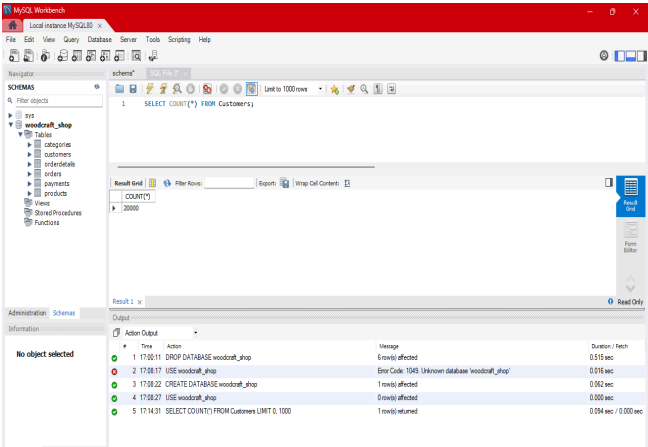
```
mysqldump -u root -p woodcraft_shop_db > backup.sql
```

Backup Steps:

- Step 1: For backup.sql we use cmd to get the backup.sql file
- Step 2: We use MySQL Workbench to recover the database.
- Step 3: Import Backup File

- Go to **Server** → **Data Import**
- Select **Import from Self-Contained File**
- Choose **/database/backup .sql**
- Set target schema: **woodcraft_shop**
- Click **Start Import**

Proof of Restoration



Indexing & Performance Improvements

```
-- Orders Table
CREATE INDEX idx_orders_customer
ON Orders(customer_id);

-- OrderDetails Table
CREATE INDEX idx_orderdetails_product
ON OrderDetails(product_id);

-- Payments Table
CREATE INDEX idx_payments_order
ON Payments(order_id);

CREATE INDEX idx_payments_date
ON Payments(payment_date);

-- Products Table
CREATE INDEX idx_products_category
ON Products(category_id);
```

Indexes were created on several tables to improve the performance and speed of database queries in the system. In the **Orders** table, an index was added on **customer_id** to make it faster to retrieve all orders belonging to a specific customer, especially when joining tables or filtering records.

In the **OrderDetails** table, an index was created on **product_id** to improve the speed when searching for products included in different orders and to support faster joins between order details and products.

For the **Payments** table, indexes were added on **order_id** and **payment_date**. The **order_id** index helps quickly match payments to their corresponding orders, while the **payment_date** index improves performance when filtering or generating reports based on payment history and dates.

In the **Products** table, an index was placed on **category_id** to make it faster to retrieve products under the same category, which is useful for filtering and displaying product lists.

Overall, these indexes help optimize query execution, reduce search time, and improve the overall efficiency of the system, especially when handling large amounts of data and multiple table relationships.

Reflection and Limitations

The development of the LikhaKahoy Woodcrafts system gave the group an opportunity to apply database concepts such as data organization, indexing, and efficient query processing in a real business context. Through this project, we gained a better understanding of how relational databases support daily operations, especially in managing products, orders, and customer information. It also helped improve our teamwork, communication, and problem-solving skills as we worked through different system requirements.



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REPUBLIC OF THE PHILIPPINES
BICOL UNIVERSITY
POLANGUI

Polangui, Albay

Email: bupc-dean@bicol-u.edu.ph



The system includes a user-friendly interface to make navigation and transactions easier; however, there are still some limitations. For instance, tracking unsold products or inactive customers requires additional queries and is not automatically displayed in the system. Security features are also limited and can still be enhanced through stronger authentication and access control mechanisms. In the future, the system can be improved by adding advanced reporting features, real-time data monitoring, and better security measures to support business growth.